Music and the Piano From the Ground Up

Keys, Scales, Chords, Inversions, Rhythm, and Sight-Reading Without Memorization

A First-Principles Reader

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Preface: Seeing the Piano Instead of Memorising It

Why so many beginners quit too early

Walk into a music shop or open a traditional beginner book and watch what gets handed to a new piano player in the first month. There is a poster of twelve major scale tables, another poster of forty-eight chord diagrams, a page of Italian speed words, and a bundle of letter rhymes such as *Every Good Boy Deserves Fudge* for counting lines one by one.

Faced with that stack of charts, most thoughtful adults draw a reasonable conclusion. Playing the piano must require memorising hundreds of disconnected facts before a single song makes sense. When life gets busy, the flashcards gather dust and the keyboard stays closed.

That conclusion is completely understandable, and completely backwards. You do not need to memorise forty-eight chord diagrams for the same reason you do not memorise a separate multiplication table for Tuesdays. A piano keyboard is a physical ruler laid out in plain sight. Unlike a guitar fretboard or a violin fingerboard, every pitch sits in strictly increasing order from left to right, one key per step, with nothing hidden inside a brass tube or behind a fret. Once you know how to read the spacing on that ruler, every scale, chord, and inversion shows itself right under your hands.

Two minutes at the keys right now

You do not need to wait until Chapter 3 to hear harmony work. If you have a piano, a small portable keyboard, or even a phone keyboard app nearby, try this two-minute experiment right now. If no instrument is near you, trace the keys with your eyes on the top strip of Figure 1.

1. Look at the raised black keys. Notice that they are not spaced evenly: they alternate in small clusters of **two black keys** (like a pair of chopsticks) and **three black keys** (like a three-pronged fork) all the way up the piano.
2. Spot any group of **two black keys** near the middle of the keyboard. Rest your right-hand thumb on the white key sitting immediately to the **left** of those two black keys. That note is called C.
3. Starting from C as 0, count four individual keys to the right — counting *every* key you pass, black and white alike $(1, 2, 3, 4)$. Your middle finger lands on the white key E, just to the right of the two black keys.

4. From E, count three more individual keys to the right $(1, 2, 3)$. Your little finger lands on the white key G. Press all three keys (C, E, and G) down together. Listen to the bright, settled ring of a C **major chord**.

5. Now keep your thumb on C and your little finger on G. Slide *only* your middle finger down one single key to the left, onto the black key $E\flat$ (the right-hand chopstick). Press those three keys together. Hear the mood turn warm, thoughtful, and inward — a C **minor chord**.

Intuition

Look at what you just did without a single flashcard. Counting +4 steps and +3 steps builds *every* major chord on the piano, starting on any of the twelve keys. Flipping that spacing to +3 steps and +4 steps — lowering the middle note by one key — turns any major chord into its minor partner. Twenty-four chords just collapsed into two hand shapes.

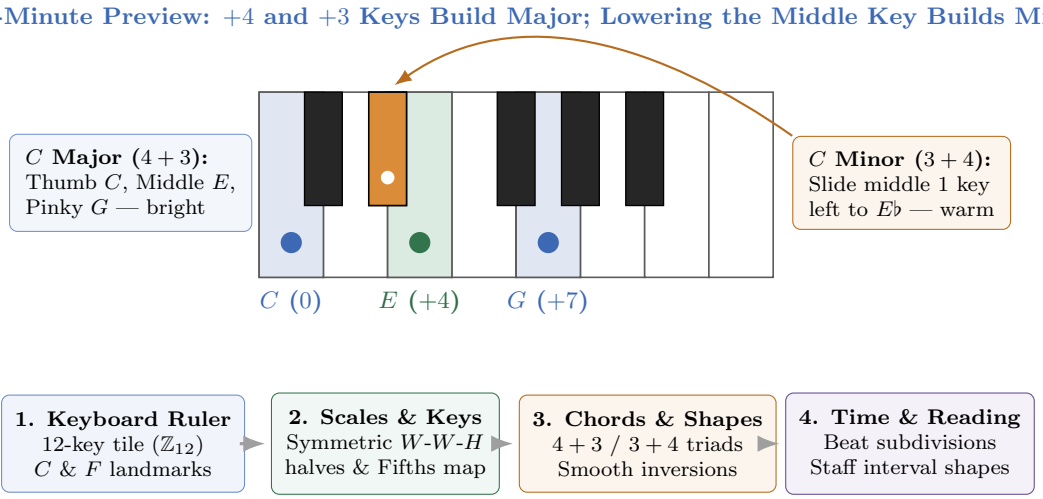


Figure 1: Top: Building C major (C–E–G) and C minor (C–$E\flat$–G) by counting individual keys from the group of two black keys. Bottom: How the four ideas in this book grow out of one twelve-key ruler.

Why music sits naturally beside mathematics

Why include a piano and music reader inside a mathematics curriculum? Because music is geometry and arithmetic that you can hear with your ears and feel under your ten fingers.

Pluck a guitar or piano string, then touch its exact midpoint: the vibrating half-string oscillates twice as fast $(2 : 1)$ and produces the same note one **octave** higher. Vibrate two-thirds of the string $(3 : 2)$ and you hear the clean, open interval called a **fifth**. Space twelve piano keys across each octave by multiplying frequency by $2^{1/12}$ at each step, and those twelve pitches wrap around a twelve-hour modular clock ($\mathbb{Z}_{12}$). On that clock, shifting a melody into a new musical key is a rigid translation ($x \mapsto x + k \bmod 12$), and inverting a chord is cyclic permutation across octaves.

You do not need any advanced mathematics to read this book. Simple counting up to twelve, plain fractions such as $\frac{3}{4}$ and $\frac{6}{8}$, and a willingness to test small patterns with your hands are plenty. Whenever a small piece of arithmetic shows up, its sole job is to save you from memorising a table.

Where we are going

This book assumes zero prior music lessons and zero background in reading sheet music. We move upward in three short, connected parts:

- **Part I: The Geometry of the Keyboard, Scales, and Keys (Chapters 1–2).** In Chapter 1 we shrink an eighty-eight-key piano down to one repeating twelve-key tile, find C and F instantly without stickers, see why sharps ($\sharp$) and flats ($\flat$) are simple left-right directions, and hear why $2:1$ and $3:2$ wave ratios sound settled. In Chapter 2 we build all twelve major scales from two matching four-note halves, find every natural minor and pentatonic scale in seconds, and read the Circle of Fifths as a neighborhood map.
- **Part II: Harmony Under Your Hands: Chords and Inversions (Chapters 3–4).** In Chapter 3 we turn chord counting into three visual keyboard shapes, walk a fixed hand claw up any scale to generate the six chords that belong to a key, and add warm seventh chords. In Chapter 4 we fold chords into smooth **inversions** so your right hand glides one key at a time instead of leaping wildly across the piano, while your left hand anchors clean bass notes.
- **Part III: Time, Rhythm, and Reading Sheet Music Effortlessly (Chapter 5).** We unpack steady pulse, time signatures ($\frac{4}{4}$, $\frac{3}{4}$, and $\frac{6}{8}$), spoken subdivisions that cure rushing on dotted notes, and a drum-pattern trick for two-hand coordination. Then we connect the eleven-line Grand Staff directly to Middle C and read melodies and chords by **landmarks and shapes** rather than spelling out letters.

What is really happening

Throughout the book, short shaded boxes pause the main explanation so each physical habit has time to settle. *Intuition* boxes highlight simple visual shortcuts on the keys. *Pause* boxes invite you to check a pattern with your fingers before turning the page. *Common trap* boxes flag habits that slow beginners down, and *Work it yourself* boxes give two-minute exercises at the piano.

Focus

One promise guides every page ahead: *never memorise what you can see directly on the keys.* Sit near a keyboard if you can, play every small example slowly with relaxed shoulders, and listen closely. Understanding how the patterns fit together makes even five minutes at the piano a quiet pleasure.

Part I

The Geometry of the Keyboard, Scales, and Keys

Chapter 1

The Twelve-Key Pattern Hiding Inside Eighty-Eight Keys

Stand in front of a full-sized grand or upright piano for the first time and your eyes meet eighty-eight black and white keys stretching nearly four feet wide. Looking at all eighty-eight keys at once feels like staring at a yardstick with eighty-eight unnumbered tick marks. Where do your hands even belong?

Take two steps back and squint slightly at the raised black keys. A calm, repeating rhythm jumps out right away:

two black keys, gap, three black keys, gap, two black keys, gap, three black keys.

An eighty-eight-key piano is not eighty-eight different notes. It is **one twelve-key tile** — seven white keys and five raised black keys — copied seven full times side by side, bookended by three extra low keys on the far left and one final high key on the far right ($3+7\times12+1=88$). Understand that single twelve-key tile once, and you understand the entire length of the instrument.

1.1 Instant Landmarks Without Stickers

Many beginners paste paper letter stickers onto their white keys. Within a week, their eyes are glued to the stickers rather than listening to the sound, and sitting down at any unlabelled piano feels like walking into a dark room. You never need stickers because the raised black keys hand you two permanent physical landmarks by sight and by touch.

Look at the two black-key clusters inside each twelve-key tile:

- **The Chopsticks (group of 2 black keys):** Slide your fingertip onto the white key sitting immediately to the **left** of any pair of two black keys. That note is always C.
- **The Fork (group of 3 black keys):** Slide your fingertip onto the white key sitting immediately to the **left** of any group of three black keys. That note is always F.

Once your eye spots C and F, the remaining five white keys fill in alphabetically from left to right

using only the first seven letters of the alphabet (A through G):

$$C \to D \to E \to F \to G \to A \to B \to C.$$

After G, the musical alphabet wraps straight back to A (never H). Notice where each white key sits in Figure 1.1. The note D sits snugly between the two chopsticks, while E sits right after them. Similarly, G and A sit inside the three-pronged fork, and B sits right after the fork just before the next C.

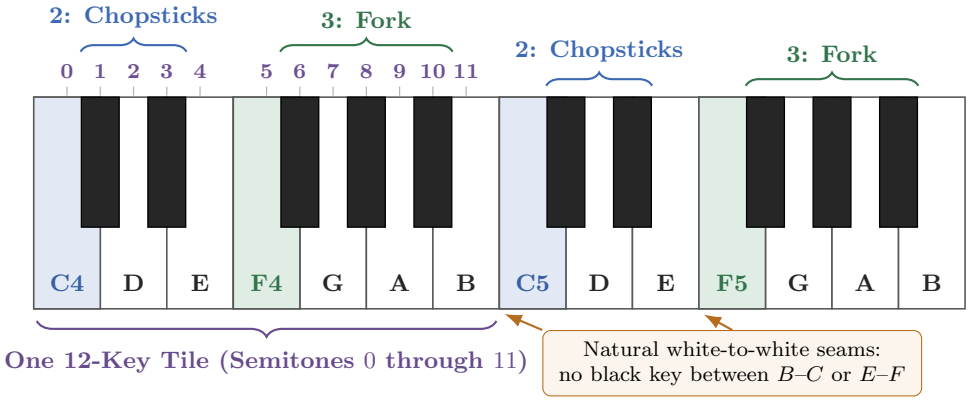


Figure 1.1: Two repeating twelve-key tiles from Middle C ($C4$) up to $B5$. Every C sits immediately left of the two black chopsticks; every F sits immediately left of the three-pronged fork. Semitone indices 0 through 11 count all twelve keys in order inside each tile.

Musicians attach a small octave number to tell one C from the next. The C closest to the middle of the piano bench (usually right underneath the brand name on the fallboard) is called **Middle** C, written $C4$. Moving right to the next C gives $C5$; moving left gives $C3$, $C2$, and $C1$, where notes rumble deep in the bass.

We also number the five fingers on *both* hands outward from the thumbs:

1	2	3	4	5
Thumb	Index finger	Middle finger	Ring finger	Little finger (pinky)

Because your hands are mirror images of each other, finger 1 is always the thumb on *both* hands, and finger 5 is always the pinky.

Work it yourself

Blindfolded landmark drill (60 seconds): Close your eyes in front of a keyboard. Lightly brush the raised black keys with fingers 2, 3, and 4 until you feel a cluster of **two** black keys. Slide your thumb down onto the smooth white key immediately to their left, play it, and open your eyes to verify that you landed on C. Repeat three times across low, middle, and high octaves, then try the same touch test for the group of **three** black keys to land on F.

1.2 Half Steps and Whole Steps: Look at the Back of the Keys

Look along the *front* edge of the piano and you see a solid row of white keys, which makes it tempting to think that stepping from one white key to the next is always the same musical distance. Now shift your gaze two inches deeper, toward the **back edge** of the keys near the wooden fallboard. Back there, narrow white tails and raised black keys line up in one even row of twelve strips per octave.

That back rail shows the true unit of distance on the piano:

- A **half step** (also called a **semitone**) moves **one physical key** to the left or right along the back rail, with *zero* keys skipped in between. Most half steps travel from a white key up to its neighbor black key (such as $C \to C\sharp$) or from a black key to a white key. However, where there is **no black key** in the gap — right across E–F and B–C — stepping directly from white key to white key is *also* one half step.
- A **whole step** (also called a **whole tone**) equals **two half steps**: you skip over **one physical key** (black or white) along the back rail. Going from C to D skips the black key $C\sharp$, so $C \to D$ is a whole step. Going from E up a whole step must skip the adjacent white key F and land on the black key $F\sharp$, as Figure 1.2 shows.

Why the technique works

Why did instrument makers leave out a black key between E–F and B–C instead of alternating white and black keys forever? Imagine an eighty-eight-key keyboard of strictly alternating white-black-white-black keys. Every key would look and feel identical to every other key; you could never spot C or F without counting from the far end of the piano. Grouping the twelve semitones into seven white keys (an ancient seven-note vocal scale) and five raised black keys gives your eyes and fingers an unmistakable map.

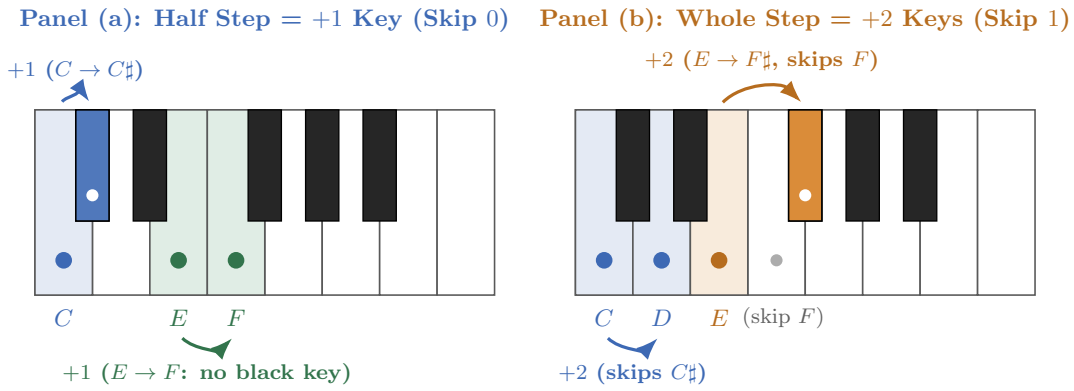


Figure 1.2: Counting along the back edge of the keys. Left: Every half step moves +1 physical key with nothing skipped — whether white-to-black ($C \to C\sharp$) or white-to-white across a missing black key ($E \to F$). Right: Every whole step moves +2 half steps, skipping one key in between ($C \to D$ skips $C\sharp$; $E \to F\sharp$ skips the white key F).

Common trap

The single most common beginner mistake at the piano is assuming that "whole step means white key to white key." Test $E \to F$ and $B \to C$ along the back edge: zero keys sit between them, so both are **half steps**. Whenever you count a whole step up from E or B, you *must* skip the neighboring white key and land on the black key $F\sharp$ or $C\sharp$.

1.3 Sharps ($\sharp$) and Flats ($\flat$) Are Directions, Not Separate Notes

Beginners sometimes treat "sharp notes" and "flat notes" as mysterious extra pitches reserved for advanced players. In truth, $\sharp$ and $\flat$ are just one-step directional arrows:

- **Sharp ($\sharp$)** means **step** 1 **half step to the right** (one physical key higher in pitch).
- **Flat ($\flat$)** means **step** 1 **half step to the left** (one physical key lower in pitch).

Look at the black key sitting between F and G. Start on F and step one half step right: you arrive at $F\sharp$. Now start on G and step one half step left: you arrive at $G\flat$ — the *exact same physical key*. Musicians call two names for the same physical key **enharmonic equivalents**. Calling one street corner "just east of 5th Avenue" ($F\sharp$) or "just west of 6th Avenue" ($G\flat$) describes the exact same patch of pavement from two neighboring landmarks. In Chapter 2 we will see how choosing between $F\sharp$ and $G\flat$ keeps scales clean so every letter from A to G appears once.

Pause

Pause and test the rule: Because $\sharp$ simply means "+1 physical key right," what key on the piano is $E\sharp$? Put your finger on E and move one physical key to the right: you land directly on the white key F. Likewise, $C\flat$ is one key left of C, which is the white key B. Sharps and flats do not mean "black keys only" — they mean +1 key and −1 key from wherever you started.

1.4 Why Music Sounds Good: Wave Ratios and the Twelve-Step Clock

Press C and G together on a piano and the two notes blend into a calm, steady ring. Press C and $C\sharp$ together right beside each other and the air buzzes with rough, restless friction. Why do certain key distances blend smoothly while others clash?

When a piano hammer strikes a steel string, the string vibrates back and forth hundreds of times per second, pushing waves of air toward your eardrum. For example, the A above Middle C ($A4$) vibrates at 440 cycles per second (440 Hz). Two notes sound settled (**consonant**) when their vibration rates lock into a **ratio of small whole numbers**, so their airwaves line up in a short, clean repeating cycle:

1. **The Octave ($2:1$ ratio, +12 half steps):** Double 440 Hz to 880 Hz ($A5$), or halve it to 220 Hz ($A3$). Two crests of the higher wave align with every single crest of the lower wave. Your ear hears the two waves fuse so cleanly that we give both keys the *same letter name* (A).

2. **The Perfect Fifth ($3:2 = 1.50$ ratio, $+7$ half steps):** Multiply a string's frequency by $\frac{3}{2}$ (for example, 220 Hz $\to$ 330 Hz, which is $A \to E$, or $C \to G$). Every three waves of the upper note lock cleanly onto two waves of the lower note, producing an open, grounded ring.
3. **The Major Third ($5:4 = 1.25$ ratio, $+4$ half steps):** Multiply frequency by $\frac{5}{4}$ (such as $C \to E$). Five upper waves align with four lower waves, adding warmth and brightness.

History

Ancient Greek, Chinese, and Arabic string makers tuned harps and lutes by stacking pure $3:2$ fifths one after another. Twelve pure fifths almost return to seven pure octaves, missing by about 1.36% because $(3/2)^{12} \approx 129.75$ whereas $2^7 = 128$ (no power of $\frac{3}{2}$ can ever equal a power of 2). For centuries, hiding that tiny leftover gap meant some keys on a harpsichord sounded sweet while distant keys howled out of tune. Dividing each $2:1$ octave into twelve *equal* multiplicative steps — **twelve-tone equal temperament**, celebrated by J. S. Bach's *Well-Tempered Clavier* (1722) and standard on modern pianos — spreads that tiny gap invisibly across all twelve keys so every key plays cleanly.

Formal

In **twelve-tone equal temperament**, stepping up one half step multiplies frequency by the constant factor

$$r = 2^{1/12} \approx 1.059463 \qquad (\text{about } +5.95\% \text{ per key}).$$

Stepping up k half steps multiplies frequency by $r^k = 2^{k/12}$:

- $k = 12$ half steps (one octave): $2^{12/12} = 2.0000$ (exact $2:1$ octave).
- $k = 7$ half steps (perfect fifth, e.g. $C \to G$): $2^{7/12} \approx 1.4983$ (within 0.11% of $\frac{3}{2} = 1.5000$).
- $k = 4$ half steps (major third, e.g. $C \to E$): $2^{4/12} \approx 1.2599$ (close to $\frac{5}{4} = 1.2500$).

Because every $+12$ steps doubles frequency and repeats the note name, the twelve pitch classes form the modular clock $\mathbb{Z}_{12} = \{0, 1, 2, \ldots, 11\}$ shown in Figure 1.3. Shifting any melody or chord up by s half steps maps every key x to $(x + s) \bmod 12$, preserving every relative ratio $2^{((x+s)-(y+s))/12} = 2^{(x-y)/12}$.

Intuition

Look at the blue triangle $(0, 4, 7)$ inside the twelve-hour clock in Figure 1.3. Turn that rigid cardboard triangle two hours clockwise (add $+2$ to every corner): $(0, 4, 7)$ lands on $(2, 6, 9)$, which is D–$F\sharp$–A, a D major chord. Because equal temperament spaces every hour on the clock by the same ratio $2^{1/12}$, turning a shape on the clock never stretches or warps its harmony.

Exercises

Exercise 1.1. Without looking at Figure 1.1, name the white key that sits (a) directly between the two chopsticks, (b) immediately to the right of the three-pronged fork, and (c) immediately to the left of the chopsticks. Check each answer on a keyboard or on Figure 1.1.

Exercise 1.2. Starting on the key B, count (a) one half step to the right, and (b) one whole step to the right. Which physical keys (white or black, and what letter names) do you land on? Now repeat

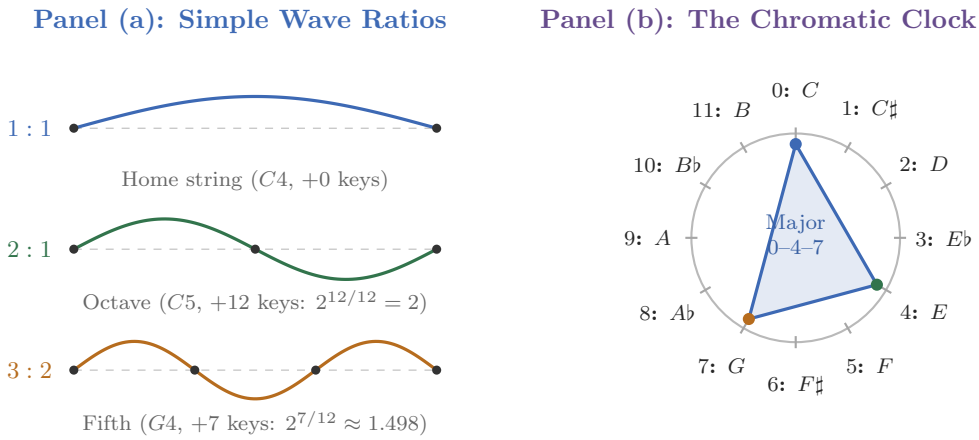


Figure 1.3: Left: Vibrating string harmonics showing why the octave ($2:1$, +12 semitones) and fifth ($3:2$, +7 semitones) align cleanly. Right: Wrapping the twelve semitones around the modular clock $\mathbb{Z}_{12}$, where C major forms the rigid triangle $(0, 4, 7)$ — rotating that same triangle clockwise builds every major chord.

both steps starting on F.

Exercise 1.3. Give two different directional names (enharmonic equivalents) for (a) the middle black key of the three-black-key fork, and (b) the white key immediately to the left of C.

Exercise 1.4. On the twelve-note clock $\mathbb{Z}_{12}$ ($0 = C, 1 = C\sharp, \ldots, 11 = B$), rotate the C major triangle $(0, 4, 7)$ clockwise by +7 hours (calculating each corner mod12). Which three note names do you land on, and what major chord did you just build?

Chapter 2

Scales and Keys Without Memorising Tables

Play every single key from Middle C up to $C5$ in order — all twelve white and black keys one after another. That twelve-note climb is called the **chromatic scale**. It sounds like a cartoon character tiptoeing up a staircase: every step is the exact same size (+1 half step), so no note feels like home.

Now play only the seven white keys from $C4$ to $C5$:

$$C \ \rightarrow \ D \ \rightarrow \ E \ \rightarrow \ F \ \rightarrow \ G \ \rightarrow \ A \ \rightarrow \ B \ \rightarrow \ C.$$

Instantly your ear hears *Do–Re–Mi–Fa–Sol–La–Ti–Do.* Even if you stop on B, your ear begs for one final step up to C to finish the sentence. Why do seven unevenly spaced notes create a clear sense of home while twelve evenly spaced notes float without gravity? Because the mix of whole steps (+2) and half steps (+1) breaks the symmetry of the twelve-hour clock, letting your ear locate "home" within two or three notes.

2.1 One Recipe for All Twelve Major Scales

Count the physical half steps between neighboring white keys as you walk from C up to C:

$$C \xrightarrow{+2} D \xrightarrow{+2} E \xrightarrow{+1} F \xrightarrow{+2} G \xrightarrow{+2} A \xrightarrow{+2} B \xrightarrow{+1} C.$$

Write W for a whole step (+2 keys) and H for a half step (+1 key). That spacing is the universal blueprint for **every major scale** on the piano:

$$W \ - \ W \ - \ H \qquad - \ W \ - \qquad W \ - \ W \ - \ H \qquad (2+2+1+2+2+2+1=12 \text{ half steps}).$$

Look at the symmetry hiding inside that seven-step string. You do not even have to remember a seven-letter list. Every major scale is simply **two matching four-note halves** ($W-W-H$), separated in the middle by one whole-step bridge (W):

- **Lower four-note half ($W-W-H$):** Start on note 1, take two whole steps and one snug half step ($C-D-E-F$).

- **Middle bridge (W):** Step up one whole step ($F \to G$) to begin the second half.
- **Upper four-note half ($W - W - H$):** Repeat the exact same two whole steps and final snug half step ($G - A - B - C$) right into the upper home note.

Formal

Starting on any root key $x_1 \in \mathbb{Z}_{12}$, the **major scale** visits the cumulative semitone offsets

$$(0,\ 2,\ 4,\ 5,\ 7,\ 9,\ 11,\ 12) \quad \text{above } x_1.$$

The lower four notes $(0, 2, 4, 5)$ and the upper four notes $(7, 9, 11, 12) = 7 + (0, 2, 4, 5)$ have the exact same internal spacing $(+2, +2, +1)$. Step 7 (at $+11$ semitones, just 1 key below the octave 12) is called the **leading tone** because its tight half-step gap pulls strongly upward into home (12).

Why does C major use purely white keys? Because when you start on C, the two H steps in the recipe fall directly onto $E \to F$ and $B \to C$, where the piano already lacks black keys.

Now watch what happens when we start that exact same $W - W - H$ recipe on two neighboring keys in Figure 2.1 — with zero tables memorised:

1. **Build G Major starting on G:**
   - *Lower half ($W - W - H$):* Start on G. Whole step to A, whole step to B, half step across the natural gap to C. That gives $G - A - B - C$ (all white keys — in fact, the exact upper half of C major).
   - *Bridge (W):* Whole step from C lands on D.
   - *Upper half ($W - W - H$):* Start on D. Whole step to E. Now take a **whole step** from E: since $E \to F$ is only a half step, you *must* skip F and land on the black key $F\sharp$. Finally, one snug half step from $F\sharp$ lands right home on G.

   Without memorising anything, keyboard geometry just told you that the key of G major has **one sharp ($F\sharp$)**. Try playing G major with a plain white F by mistake and your ear will wince right at the top: $F \to G$ is a clunky whole step that loses the crisp half-step pull home.
2. **Build F Major starting on F:**
   - *Lower half ($W - W - H$):* Start on F. Whole step to G, whole step to A. Now take a **half step** from A: one physical key right of A is the rightmost black key of the three-key fork, $B\flat$.
   - *Bridge (W) and upper half ($W - W - H$):* Whole step from $B\flat$ skips B and lands on C; from C, the familiar white-key run $C - D - E - F$ finishes $W - W - H$ home to F.

   Again with zero flashcards, the ruler showed you that F major has **one flat ($B\flat$)**.

Common trap

Why must we call the fourth note of F major $B\flat$ rather than $A\sharp$, even though both names point to the same black key? Because of the **One-Letter-Per-Step Rule**: every seven-note major or minor scale uses each letter A, B, C, D, E, F, G *once and only once* in alphabetical order. Writing $F - G - A - B\flat - C - D - E$ gives every note its own unique letter (and its

own distinct line or space on sheet music). Writing $F - G - A - A\sharp - C$ would repeat the letter A twice and skip B altogether.

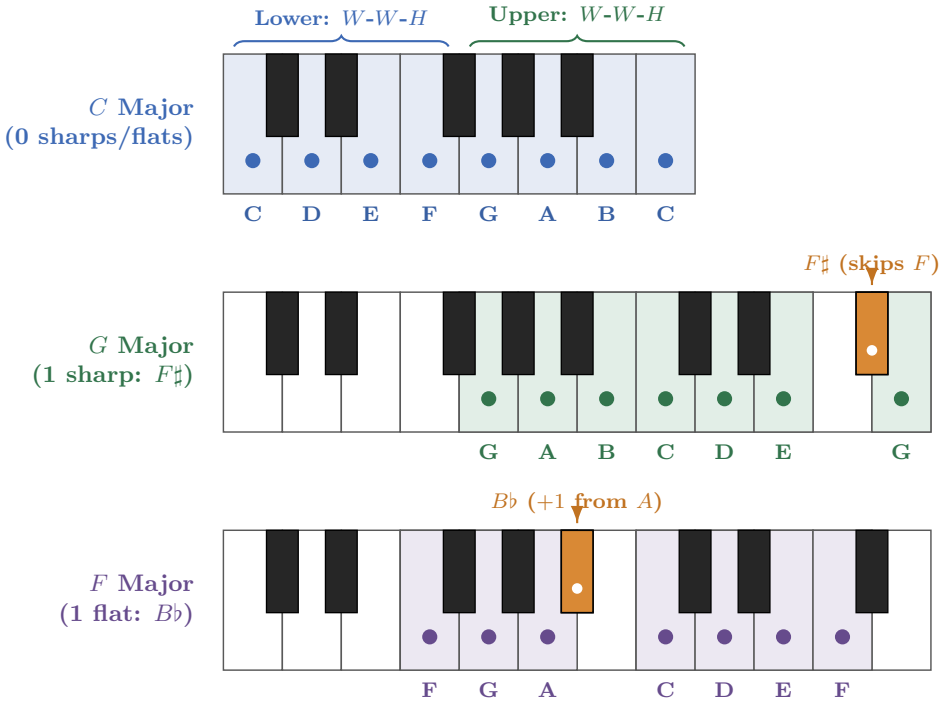


Figure 2.1: Three major scales aligned vertically on the same keyboard strip ($C4$ to $G5$). Every scale follows $W - W - H$ (lower four notes), a whole-step bridge W, and $W - W - H$ (upper four notes). Starting on G forces $F\sharp$ at step 7; starting on F forces $B\flat$ at step 4.

2.2 How Five Fingers Play Eight Notes Smoothly: The Thumb Tunnel

You have five fingers on your right hand, yet a full scale spans eight notes. If you play $C-D-E-F-G$ with fingers $1-2-3-4-5$, you run out of fingers on G and have to hop your whole hand awkwardly sideways.

Pianists solve that problem with one gentle motion called the **thumb tunnel**:

- **Going up in the right hand ($C \to C$):** Play $C - D - E$ with fingers $\mathbf{1} - \mathbf{2} - \mathbf{3}$. As finger 3 plays E, let your thumb (**1**) glide quietly underneath the natural arch of your palm onto F. Now your five fingers sit naturally over $F - G - A - B - C$: walk $\mathbf{1} - \mathbf{2} - \mathbf{3} - \mathbf{4} - \mathbf{5}$ smoothly to the top.
- **Coming back down ($C \to C$):** Walk five fingers down $C-B-A-G-F$ using $\mathbf{5} - \mathbf{4} - \mathbf{3} - \mathbf{2} - \mathbf{1}$. While your thumb holds F, pivot finger **3** smoothly *over* your thumb onto E, then finish $E - D - C$ with $\mathbf{3} - \mathbf{2} - \mathbf{1}$.
- **Left hand mirror ($C \to C$):** Going up from $C3$ to $C4$, walk all five fingers $\mathbf{5} - \mathbf{4} - \mathbf{3} - \mathbf{2} - \mathbf{1}$ on $C-D-E-F-G$, pivot finger **3** over the thumb onto A, and finish $\mathbf{3} - \mathbf{2} - \mathbf{1}$ on $A-B-C$.

Intuition

Keep your wrist level and your elbow relaxed near your side during the thumb tunnel. Avoid kicking your elbow outward like a chicken wing when your thumb passes under finger 3; let your thumb move horizontally under a soft, curved hand arch, as if holding a small tangerine under your palm.

2.3 Natural Minor and the Five-Note Pentatonic Playground

Must you learn a brand-new step formula to play in minor keys? Not at all. Every major scale already contains its partner **natural minor scale** using the *exact same set of keys*.

Put your finger on any major home note and slide 3 **half steps to the left** (skipping two keys, landing on step 6 of that major scale):

- From C, slide 3 half steps left ($C \to B \to B\flat \to A$): you land on A. Play the exact same white keys of C major starting and ending on A:

  $$A \ \to \ B \ \to \ C \ \to \ D \ \to \ E \ \to \ F \ \to \ G \ \to \ A \qquad (W - H - W - W - H - W - W).$$

  That is A **natural minor**, the **relative minor** of C major.
- From G major (1 sharp: $F\sharp$), slide 3 half steps left to E: playing those same keys from E to E ($E - F\sharp - G - A - B - C - D - E$) gives E **natural minor**.
- From F major (1 flat: $B\flat$), slide 3 half steps left to D: playing from D to D ($D - E - F - G - A - B\flat - C - D$) gives D **natural minor**.

Why does A–B–C–D–E–F–G–A sound pensive and minor even though it uses the exact same white keys as bright C major? Because your ear measures every note against the *home bass note*. In C major, step 3 (E) sits +4 half steps above C (a bright major third). In A minor, step 3 (C) sits only +3 half steps above A (a darkened minor third). Note that in A natural minor, step 7 (G) sits a whole step below A. When composers want a stronger pull back home to A, they often raise step 7 by one half step from G to $G\sharp$ (forming A **harmonic minor**), creating a tight $G\sharp \to A$ half-step magnet.

The five-note pentatonic scale: instant melody without clashes

Look back at the seven notes of C major ($C - D - E - F - G - A - B$) and ask which two notes create half-step tension. They are step 4 (F, rubbing against E) and step 7 (B, rubbing against C). Lift those two half-step neighbors out of the scale and five open, friendly notes remain:

$$\text{Major pentatonic (steps } 1, 2, 3, 5, 6\text{):} \qquad C \ - \ D \ - \ E \ - \ G \ - \ A.$$

Start that same five-key set on A ($A - C - D - E - G$) and you get A **minor pentatonic**, the backbone of folk, blues, rock, and pop melodies worldwide. Best of all, shift $C - D - E - G - A$ up +6 half steps onto $F\sharp$ ($G\flat$) and look at Figure 2.2: those five notes become **the five black keys on the piano** ($G\flat - A\flat - B\flat - D\flat - E\flat$ for major pentatonic, or $E\flat - G\flat - A\flat - B\flat - D\flat$ for minor pentatonic). Because zero half steps exist anywhere among the five black keys, literally *any* note order you improvise on the black keys sounds melodic right away.

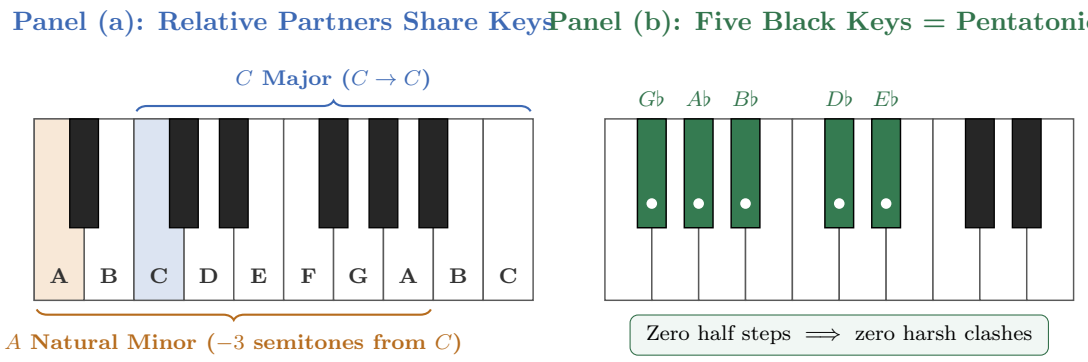


Figure 2.2: Left: Sliding 3 half steps down from any major tonic ($C \to A$) gives its relative natural minor on the exact same physical keys. Right: All five raised black keys ($G\flat, A\flat, B\flat, D\flat, E\flat$) form a complete major pentatonic scale (or $E\flat$ minor pentatonic when started on $E\flat$).

2.4 The Circle of Fifths as a Neighborhood Map

Look back at C major (0 sharps) and G major (1 sharp, $F\sharp$) in Figure 2.1. Why do they share six out of seven notes? Because G sits a fifth (+7 half steps) above C, the upper four-note half of C major ($G - A - B - C$) becomes the *lower* four-note half of G major unchanged. Only one note has to rise ($F \to F\sharp$) to form the new leading tone below G.

Repeat that +7-semitone hop again from G to D: G major's upper half ($D - E - F\sharp - G$) becomes D major's lower half (keeping $F\sharp$), and the new seventh step rises from C to $C\sharp$, giving D major **two sharps ($F\sharp, C\sharp$)**.

Why does jumping by +7 on a 12-hour clock visit *all twelve* keys before repeating C? Because $\gcd(7, 12) = 1$: seven and twelve share no common factor larger than 1, so repeatedly adding +7 (mod 12) visits every single key once before returning to 0 after twelve steps ($12 \times 7 = 84 = 7 \times 12$). Laying those twelve fifth-steps around a dial produces the famous **Circle of Fifths** in Figure 2.3:

- **Clockwise (+7 semitones, up a fifth):** Each step adds **one new sharp** (always the new seventh step, a half step below the new major key):

  $$C\ (0) \to G\ (1\sharp) \to D\ (2\sharp) \to A\ (3\sharp) \to E\ (4\sharp) \to B\ (5\sharp) \to F\sharp\ (6\sharp).$$

  The sharps accumulate in fifths too: $F\sharp, C\sharp, G\sharp, D\sharp, A\sharp, E\sharp$.

- **Counterclockwise ($-7 \equiv +5$ semitones, up a fourth):** Each step adds **one new flat** (always the new fourth step):

  $$C\ (0) \to F\ (1\flat) \to B\flat\ (2\flat) \to E\flat\ (3\flat) \to A\flat\ (4\flat) \to D\flat\ (5\flat) \to G\flat\ (6\flat),$$

  accumulating flats in exact reverse order: $B\flat, E\flat, A\flat, D\flat, G\flat, C\flat$. At 6 o'clock on the dial, $F\sharp$ major ($6\sharp$) and $G\flat$ major ($6\flat$) press the *exact same piano keys*.

What is really happening

When printed sheet music places sharps or flats right after the clef at the start of every line (the **key signature**), two quick shortcuts read the Major key without counting around the

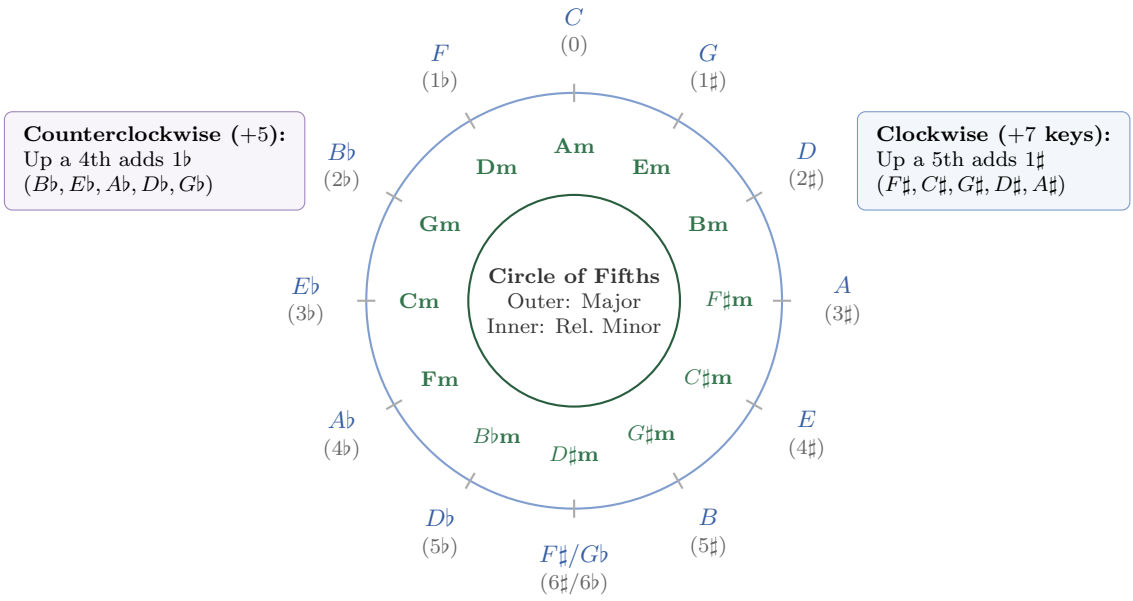


Figure 2.3: The Circle of Fifths as a neighborhood map. Adjacent keys on the dial share six out of seven notes, differing by just one sharp or flat. Every inner-ring natural minor key sits 3 half steps below its outer-ring major partner and shares the exact same key signature.

dial:

1. **Sharp key signatures:** Look at the *last* (rightmost) sharp and step **up** 1 **half step**. That is the Major key (because the newest sharp is always step 7, the leading tone). Example: three sharps $F\sharp, C\sharp, G\sharp$ — one half step above $G\sharp$ is A **major**.

2. **Flat key signatures:** One flat ($B\flat$) is F **major**. For two or more flats, the **second-to-last flat** *is* the name of the Major key directly. Example: three flats $B\flat, E\flat, A\flat$ — the second-to-last flat is $E\flat$ **major**.

If the song centers and ends on the note 3 half steps *below* that major tonic, it is in the relative minor key instead.

Work it yourself

Two-minute improvisation at the piano: Hold down the low black key $E\flat 2$ (the right-hand chopstick in the bass) with your left hand. With your right hand, play slow, unhurried phrases using *only* the five black keys in any order, leaving breathing space between short three- or four-note ideas. Now switch your left-hand bass note up three half steps to $G\flat 2$ (the leftmost prong of the three-key fork) while keeping the exact same five black keys in your right hand. Listen to the mood shift immediately from pensive $E\flat$ minor pentatonic to warm, bright $G\flat$ major pentatonic.

Exercises

Exercise 2.1. Starting on D, apply the $W - W - H$ — W — $W - W - H$ spacing step by step on a keyboard (or Figure 2.1) to write out all eight notes of the D major scale. Which two black keys does the spacing force you to play?

Exercise 2.2. Starting on $B\flat$ (the rightmost black key of the three-key fork), walk $W - W - H$ — W — $W - W - H$ and apply the One-Letter-Per-Step Rule to spell out all seven distinct notes of the $B\flat$ major scale. Verify your flat count against Figure 2.3.

Exercise 2.3. Find the relative natural minor home note (3 half steps down) for (a) D major, (b) $B\flat$ major, and (c) $E\flat$ major. Which note would you raise by +1 half step in D minor if you wanted a tight half-step leading tone into D?

Exercise 2.4. Use the two-second key-signature shortcuts to name the Major key (and its relative minor partner) for: (a) four sharps ($F\sharp, C\sharp, G\sharp, D\sharp$), and (b) four flats ($B\flat, E\flat, A\flat, D\flat$).

Part II

Harmony Under Your Hands: Chords and Inversions

Chapter 3

Chords by Hand Shape: Major, Minor, Seventh, and Key Families

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A scale plays notes one after another in a horizontal line, building a **melody**. Press three or more notes down *at the same time* and they fuse vertically into a **chord**, creating **harmony** — the emotional lighting underneath a melody. Sing the single note G over a C major chord, then over an A minor chord, and finally over an F major chord. Your sung note never moved, yet its mood shifts from bright certainty to quiet reflection to open longing.

Beginners often buy thick chord dictionaries listing hundreds of dot diagrams. None of that memorisation is needed. Almost every song you hear on the piano grows out of simple three-note stacks called **triads**, built either by counting two small half-step gaps or by spotting three visual shapes on the keys.

3.1 The $4+3$ and $3+4$ Rule: Major and Minor as a Single-Key Toggle

Sit at any key on the piano — white or black — and call that starting note your **root** (0). Rest your right-hand thumb (**1**), middle finger (**3**), and pinky (**5**) over the keys:

- **Major triad ($4+3$ half steps):** From your root (0), count +4 **half steps** right for your middle note (the **third**) and +3 **more half steps** right (+7 total from the root) for your top note (the **fifth**). Try it on C: +4 keys lands on E, and +3 more lands on G. Press $C-E-G$ together.
- **Minor triad ($3+4$ half steps):** Keep your thumb on the root (0) and your pinky on the fifth (+7). Slide *only* your middle finger **down** 1 **half step to the left** so the spacing flips to +3 **half steps** on the bottom and +4 **half steps** on top. On C, E slides left one key to $E\flat$, giving $C-E\flat-G$.

Test that one-key middle switch on two more roots right away:

- Start on D: +4 half steps lands on the black key $F\sharp$, and +3 more lands on A, giving D **major** ($D-F\sharp-A$). Lower the middle finger one key from $F\sharp$ to white F and you hold D **minor**

$(D - F - A)$.

- Start on A: $+4$ half steps lands on the black key $C\sharp$, and $+3$ more lands on E, giving A **major** $(A - C\sharp - E)$. Lower the middle finger one key from $C\sharp$ to white C and you hold A **minor** $(A - C - E)$.

Figure 3.1 shows why major and minor sound like two moods of the same home note. The outer frame of $+7$ half steps (the clean $3:2$ perfect fifth from Chapter 1) never budges, anchoring the chord firmly. Only that single middle key moves, choosing bright major $(+4)$ or warm minor $(+3)$.

Formal

Starting on any root key $r \in \mathbb{Z}_{12}$, the four foundational three-note triads are determined completely by their cumulative semitone offsets $(0, a, a+b)$ above r:

Triad Type	Symbol	Half-Step Gaps	Offsets Above Root r
Major triad	C	$4+3$	(0, 4, 7)
Minor triad	Cm	$3+4$	(0, 3, 7)
Diminished triad	C° (or Cdim)	$3+3$	(0, 3, 6)
Suspended fourth	Csus4	$5+2$	(0, 5, 7)

Adding $+s$ (mod 12) shifts any triad recipe cleanly onto any of the twelve keys.

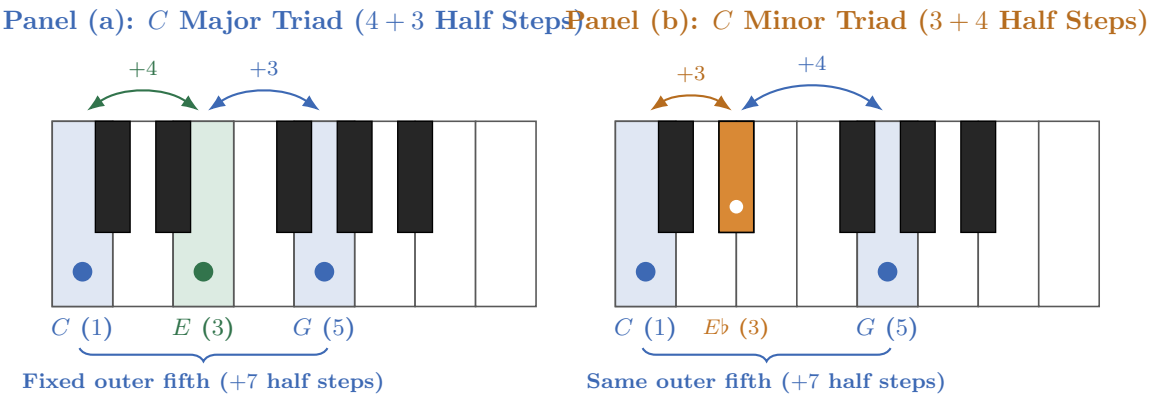


Figure 3.1: Major $(4+3)$ versus minor $(3+4)$ on the keyboard. Thumb (1) and pinky (5) hold the exact same grounding fifth (C–G, $+7$ semitones), while middle finger (3) toggles one physical key between E (major) and $E\flat$ (minor).

What about the other two rows in that table?

- **Suspended chords (sus4 $= 5+2$ and sus2 $= 2+5$):** Lift your middle finger off the third completely so the chord is neither major nor minor. Drop finger **4** onto step 4 $(C - F - G$, $5+2)$ for C**sus4**, or drop finger **2** onto step 2 $(C - D - G$, $2+5)$ for C**sus2**. Play Csus4 $\to C$ $(C - F - G$ resolving down to $C - E - G)$ and you will recognize the warm "sigh" at the end of hundreds of hymns and pop songs.
- **Diminished triad $(3+3)$:** Stack two minor thirds $(0, 3, 6)$, such as $B - D - F$. Because its outer frame shrinks from a settled 7-semitone fifth to a restless 6-semitone **tritone**, a diminished chord leans strongly inward, asking to resolve to a neighboring stable chord $(B - D - F \to C - E)$.

3.2 Three Visual Keyboard Shapes for Twelve Major Chords

Counting +4 and +3 always works, yet experienced pianists do not count to four while playing a song. Once you build the twelve major triads once, you notice a striking geometric shortcut shown in Figure 3.2: **nine of the twelve major chords fall into just three symmetric keyboard profiles**:

1. **Group 1 — All White Keys ($W-W-W$, flat row): C, F, and G major** ($C-E-G$, $F-A-C$, $G-B-D$). Your three fingers sit on one level white plane, skipping one white key between each finger.
2. **Group 2 — White–Black–White ($W-B-W$, raised triangle): D, E, and A major** ($D-F\sharp-A$, $E-G\sharp-B$, $A-C\sharp-E$). Thumb and pinky sit on white keys while your middle finger rests comfortably up on a raised black peak.
3. **Group 3 — Black–White–Black ($B-W-B$, dipped valley): $D\flat$, $E\flat$, and $A\flat$ major** ($D\flat-F-A\flat$, $E\flat-G-B\flat$, $A\flat-C-E\flat$) — notice these are simply the flat versions of D, E, A from Group 2. Thumb and pinky sit high on black keys while your middle finger dips down onto a white key in the center (often nicknamed the "cookie" shape: dark wafer, white cream, dark wafer).

What about the final three keys out of twelve? They are just as memorable:

- **$G\flat$ (or $F\sharp$) major** sits entirely on three black keys ($B-B-B$: $G\flat-B\flat-D\flat$), as flat and level as C major.
- **$B\flat$ major** ($B-W-W$: $B\flat-D-F$) and **B major** ($W-B-B$: $B-D\sharp-F\sharp$) are exact mirror twins of one another.

Intuition

Look at the payoff of those visual groups. Spot D, E, or A? Raise the middle finger onto a black key ($W-B-W$) for major, or drop all three onto white keys ($W-W-W$) for Dm, Em, and Am. Spot $D\flat$, $E\flat$, or $A\flat$? Hold the $B-W-B$ valley for major. Twenty-four major and minor chords become physical hand feels rather than written drills.

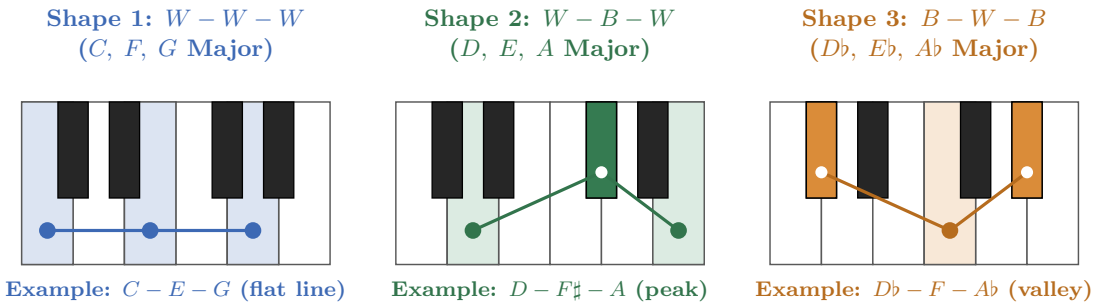


Figure 3.2: Nine of the twelve major chords belong to three unmistakable hand profiles: all-white flat line (C, F, G), white-black-white raised peak (D, E, A), and black-white-black valley ($D\flat, E\flat, A\flat$).

3.3 Why Six Chords Belong Together in Every Key

Suppose a friend sits down to write a song in the key of C major. Out of all twenty-four major and minor chords on the piano, how do songwriters immediately know which chords sound like family members that belong together?

Place your right-hand fingers $\mathbf{1-3-5}$ on $C-E-G$ (*play a white key, skip a white key, play, skip, play*). Now freeze that "play-skip-play-skip-play" claw shape and walk your whole hand up the C major scale one white key at a time, staying strictly inside the scale's notes. Because white keys mix whole steps and half steps unevenly, the keyboard automatically hands you a balanced family of major and minor chords (called **diatonic chords**), numbered with Roman numerals:

1. Step 1 ($C-E-G$, gaps $4+3$): I $=$ C **major** (uppercase Roman numeral = Major).
2. Step 2 ($D-F-A$, gaps $3+4$): ii $=$ Dm (lowercase Roman numeral = Minor).
3. Step 3 ($E-G-B$, gaps $3+4$): iii $=$ Em.
4. Step 4 ($F-A-C$, gaps $4+3$): IV $=$ F **major**.
5. Step 5 ($G-B-D$, gaps $4+3$): V $=$ G **major**.
6. Step 6 ($A-C-E$, gaps $3+4$): vi $=$ Am (the relative minor home).
7. Step 7 ($B-D-F$, gaps $3+3$): vii$^\circ$ $=$ B° (diminished).

Because every major scale has the exact same $W-W-H-W-W-W-H$ spacing, **every major key on earth produces that exact same Roman-numeral pattern**:

$$\underbrace{\text{I, IV, V}}_{\text{3 Major chords}} \quad \text{and} \quad \underbrace{\text{ii, iii, vi}}_{\text{3 Minor chords}} \quad (\text{plus vii}^\circ).$$

Harmonic gravity: Home, Stepping Out, and Tension Pulling Home

Even better, those six chords play three clear storytelling roles shown in Figure 3.3:

- **Home (Tonic: I, or vi for a softer mood):** Settled and at rest. A song feels finished when it lands on I (C major) — or bittersweet and reflective when it rests on vi (Am, which shares two notes C and E with C major).
- **Stepping Out (Subdominant: IV and ii):** Gentle forward travel away from home. Moving from C (I) to F (IV) or Dm (ii) lifts the story onto open road without harsh urgency.
- **Tension Pulling Home (Dominant: V, and vii$^\circ$):** Notice that G major ($G-B-D$) contains the leading tone B, sitting just one half step below C. Playing V builds forward expectation that resolves cleanly back into I.

Look how four of the most famous chord loops in music history read in Roman numerals, and how effortlessly they translate from C major to G major:

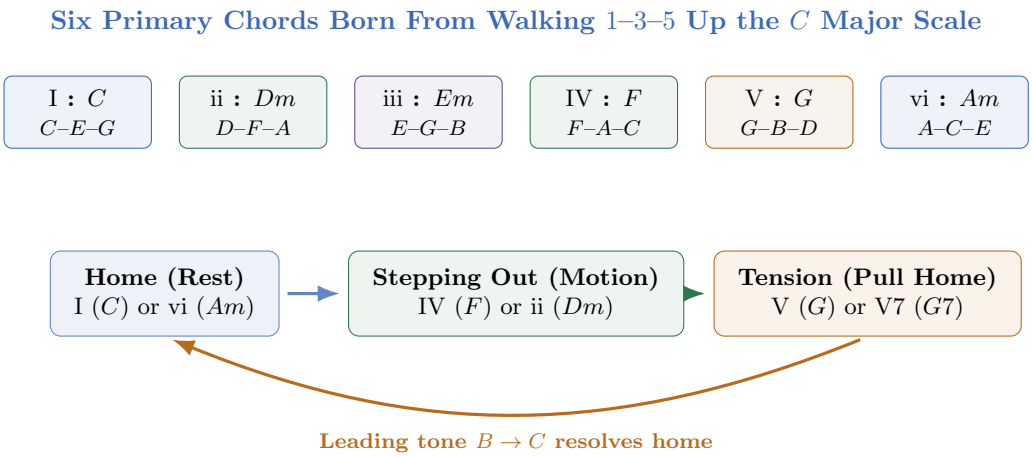


Figure 3.3: Top: Walking a 1–3–5 hand shape up the C major scale produces three major chords (I, IV, V) and three minor chords (ii, iii, vi). Bottom: Harmonic gravity moves outward from Home (I, vi) to Stepping Out (IV, ii) to Dominant Tension (V, V7), which pulls back Home.

Progression Pattern	**Character**	**In C Major**	**In G Major**
I – V – vi – IV	Anthemic pop ballad	$C-G-Am-F$	$G-D-Em-C$
vi – IV – I – V	Reflective story loop	$Am-F-C-G$	$Em-C-G-D$
ii – V – I	Classic jazz & soul cadence	$Dm-G-C$	$Am-D-G$
I – vi – IV – V	1950s doo-wop & soul	$C-Am-F-G$	$G-Em-C-D$

3.4 Four-Note Seventh Chords Made Simple

Once three-note triads $(1-3-5)$ feel natural, skip one more scale step and add a fourth note on top $(1-3-5-7)$. That four-note stack is called a **seventh chord**. Instead of memorising four-note grids, notice that every seventh chord is simply **a familiar triad in your right hand over a bass root in your left hand**:

1. **Major Seventh (Cmaj7 $= C-E-G-B$, gaps $4+3+4$):** Built naturally on step I (and IV). Cover up the bottom C with your thumb and look at the top three notes $E-G-B$: that is an E **minor triad sitting over a C bass note**. Play C in your left hand and Em in your right hand to hear the warm, unhurried glow of jazz ballads and lo-fi piano.

2. **Minor Seventh (Dm7 $= D-F-A-C$, gaps $3+4+3$):** Built naturally on steps ii, iii, and vi. Cover up the bottom D and look at $F-A-C$: Dm7 is simply an F **major triad over a D bass note** (likewise, Am7 $= A-C-E-G$ is a C major triad over an A bass).

3. **Dominant Seventh (G7 $= G-B-D-F$, gaps $4+3+3$):** Built naturally on step V (a plain G major triad plus the white key F on top, 2 half steps below the upper G). Notice what lives inside $G-B-D-F$: the note B wants to step *up* one half step to C, while F wants to step *down* one half step to E. Together B and F form a 6-semitone tritone magnet that pulls G7 irresistibly home into C major.

Pause

Pause and spot the label difference: Never mix up **Cmaj7** ($C-E-G-B$, top note 1 half step below C) with **C7** ($C-E-G-B\flat$, top note 2 half steps below C). Plain "7" with no "maj" means the bluesy, tension-filled *dominant* seventh ($4+3+3$), which pulls toward F major. Adding "maj7" keeps the dreamy scale seventh ($4+3+4$) at rest.

Work it yourself

Play a complete jazz-pop turnaround in two minutes: In your left hand, play single low bass notes $D2 \to G2 \to C2$ slowly (four steady counts per note). Above them in your right hand, play $F-A-C$ over $D2$ (making Dm7), then $F-G-B$ over $G2$ (making G7), then $E-G-B$ over $C2$ (making Cmaj7). Notice how few keys your right hand moved to produce rich, four-note studio harmony — a preview of smooth inversions in Chapter 4.

Exercises

Exercise 3.1. Count +4 and +3 half steps starting on E, and then starting on $A\flat$ (the middle black key of the three-key fork), to spell out the E major and $A\flat$ major triads. Check which visual shape group in Figure 3.2 each chord belongs to. Then slide each middle note down 1 half step to spell out Em and $A\flat m$.

Exercise 3.2. Spell out Dsus4 ($5+2$ half steps above D) and Dsus2 ($2+5$ half steps above D), and compare both with D major ($D-F\sharp-A$).

Exercise 3.3. Walk the $1-3-5$ scale claw up the F major scale ($F, G, A, B\flat, C, D, E$) to list all six primary chords (I, ii, iii, IV, V, vi) in the key of F major. Translate the progression $\mathrm{I}-\mathrm{V}-\mathrm{vi}-\mathrm{IV}$ into F major.

Exercise 3.4. Using the triad-over-bass shortcut, which right-hand major triad played over an E bass note produces Em7 ($E-G-B-D$)? And which two notes inside D7 ($D-F\sharp-A-C$) resolve by single half steps into G major ($G-B-D$)?

Chapter 4

Inversions and Voice Leading: Moving Fingers Inches, Not Feet

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Watch a beginner try to play a simple three-chord song — C major to F major to G major — using only the root-position shapes from Chapter 3. They press $C - E - G$, stare down in mild alarm, launch their whole arm four keys to the right to grab $F - A - C$, launch again to $G - B - D$, and leap all the way back down to $C - E - G$. Every chord change feels like jumping across stepping stones in the dark, and the steady beat breaks apart.

Now watch an experienced pianist play those exact same three chords. Their right hand barely moves an inch. Their wrist stays quiet, they never even glance down at the keyboard, and the chords melt smoothly into one another like a four-part choir. What are they doing differently? They are folding the chords into **inversions**.

4.1 Three Shapes of Every Chord, One Gap Detector

Remember from Chapter 1 that shifting any note up or down by +12 half steps (one octave, a $2:1$ wave ratio) keeps the *exact same pitch name*. A chord is defined by its set of pitch names, regardless of which note sits on the bottom. Whenever you take the lowest note of a three-note chord and flip it +12 half steps up to the top, you create an **inversion** of that chord.

Follow C major (C, E, G) across its three hand shapes in Figure 4.1:

1. **Root position ($C4 - E4 - G4$, right-hand fingers $\mathbf{1 - 3 - 5}$):** The root C sits at the bottom. Both gaps are small thirds (1 skipped white key in each gap), spanning five white keys total.

2. **First inversion ($E4 - G4 - C5$, right-hand fingers $\mathbf{1 - 2 - 5}$):** Flip bottom $C4$ up one octave to $C5$. Now $E4 - G4$ is a snug third at the bottom (1 skipped white key), while $G4 - C5$ opens into a wider **fourth** on top (2 skipped white keys, 5 half steps). Notice the right-hand fingering $\mathbf{1 - 2 - 5}$: placing your index finger (**2**) on $G4$ leaves a relaxed, natural stretch between finger 2 and finger 5 across the wider top gap.

3. **Second inversion ($G4 - C5 - E5$, right-hand fingers $\mathbf{1 - 3 - 5}$):** Flip bottom $E4$ up one

octave to $E5$. Now the wider fourth ($G4 - C5$, 2 skipped white keys) sits at the bottom — right where the flexible web between your thumb (**1**) and middle finger (**3**) opens effortlessly — and the snug third ($C5 - E5$) sits on top. Flip $G4$ up one more octave and you return to root position ($C5 - E5 - G5$).

Formal

Every three-note triad $\{r, t, f\}$ has exactly three close-position voicings within an octave:

- **Root position:** $(r,\ t,\ f)$ — spans 7 half steps (a fifth); right-hand fingering $\mathbf{1-3-5}$ (left hand $\mathbf{5-3-1}$).
- **First inversion:** $(t,\ f,\ r+12)$ — spans 8 or 9 half steps (a sixth), with a 5-semitone fourth on **top**; right-hand fingering $\mathbf{1-2-5}$ (left hand $\mathbf{5-3-1}$).
- **Second inversion:** $(f,\ r+12,\ t+12)$ — spans 8 or 9 half steps (a sixth), with a 5-semitone fourth on **bottom**; right-hand fingering $\mathbf{1-3-5}$ (left hand $\mathbf{5-2-1}$).

The Gap Detector: name any inverted chord in one glance

Suppose your fingers land on an unfamiliar inverted triad such as $A-D-F\sharp$, and you want to know its root name right away without flipping notes in your head. Use the **Gap Detector**:

> *Look for the wider gap (5 half steps, skipping two scale steps instead of one). The note sitting immediately **above** that wider gap is **always the root** of the chord.*

Check it on $A-D-F\sharp$: between A and D sits a wide 5-semitone fourth (B and $C\sharp$ skipped), whereas D to $F\sharp$ is a snug 4-semitone third. Which note sits directly *above* the wide gap? D — so $A-D-F\sharp$ is a D **major chord** (in second inversion). Check $E-G-C$: the wide gap sits between G and C, and the note right above that gap is C (C major, first inversion). If neither gap is wider, the chord is already in root position and the bottom note is the root.

4.2 Smooth Voice Leading: The Anchor-Note Law

Think of the three notes in your right-hand chord as three singers standing side by side. No real singer leaps five scale steps up and down on every bar when a note one step away belongs to the next chord. Moving each individual note along the shortest possible path from chord to chord is called **smooth voice leading**, and one physical habit makes it automatic at the piano:

Intuition

The Anchor-Note Law: When moving from one chord to the next, *keep any shared note frozen under the finger already holding it*, and step your remaining fingers by just one key to the nearest notes of the new chord.

Walk through a complete song progression around Middle C and watch Figure 4.2 compare clunky root-position jumps against smooth anchored inversions:

1. **Start on C major ($C4 - E4 - G4$, root position, fingers $1-3-5$).**

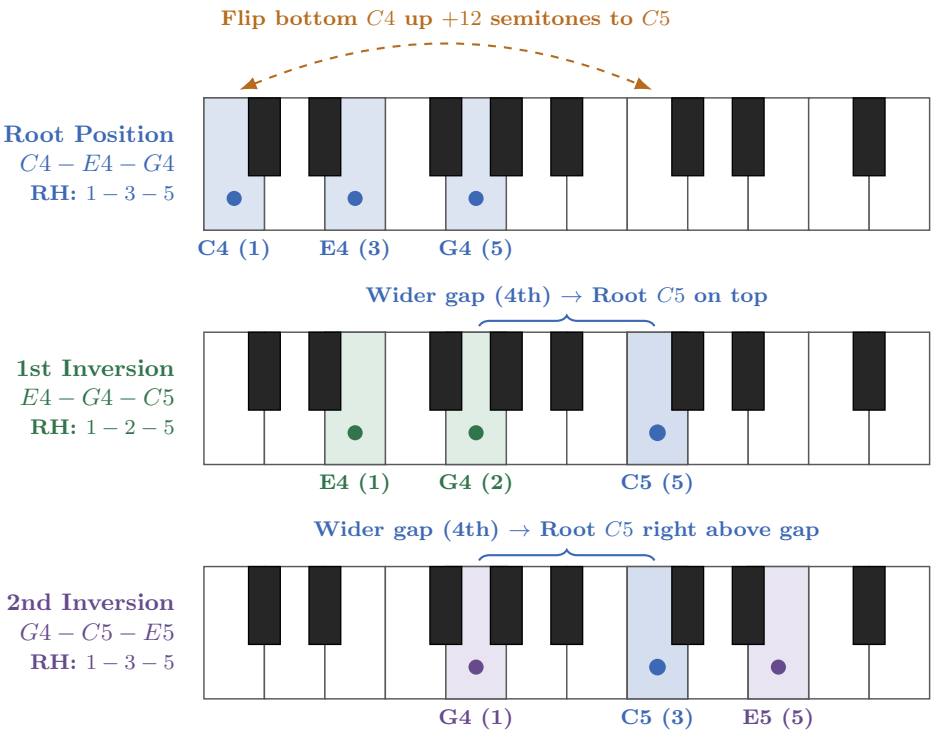


Figure 4.1: Folding C major across two octaves ($C4$ to $G5$). Flipping the bottom note up +12 semitones moves from root position $(1-3-5)$ to first inversion $(1-2-5)$ to second inversion $(1-3-5)$. In every inversion, the root ($C5$, shaded blue) sits immediately above the wider fourth gap.

2. **Move $C \to F$ major (F, A, C):** Which note do C major and F major share? $C4$. Freeze your thumb (**1**) on $C4$, and step fingers **3** and **5** up just *one white key* from $E4 - G4$ to $F4 - A4$. You are now holding $C4 - F4 - A4$ (F major, second inversion, fingers $1 - 3 - 5$). Total arm jump: zero inches.

3. **Move $C \to G$ major (G, B, D):** Return to $C4 - E4 - G4$. Which note do C major and G major share? $G4$. Freeze your pinky (**5**) on $G4$, and step your bottom two notes down just *one white key* onto $B3 - D4$ using fingers $\mathbf{1 - 2 - 5}$. You are now holding $B3 - D4 - G4$ (G major, first inversion).

4. **Move $G \to Am \to F \to C$:** From G ($B3 - D4 - G4$), step all three voices smoothly up one white key to $C4 - E4 - A4$ (A minor, first inversion, $1 - 2 - 5$). Now move from Am ($C4 - E4 - A4$) to F ($C4 - F4 - A4$): *two* shared notes ($C4$ and $A4$) stay completely still while the middle note moves a single half step ($E4 \to F4$). Finally step $F4 - A4$ down one white key to $E4 - G4$ and you are back home on C.

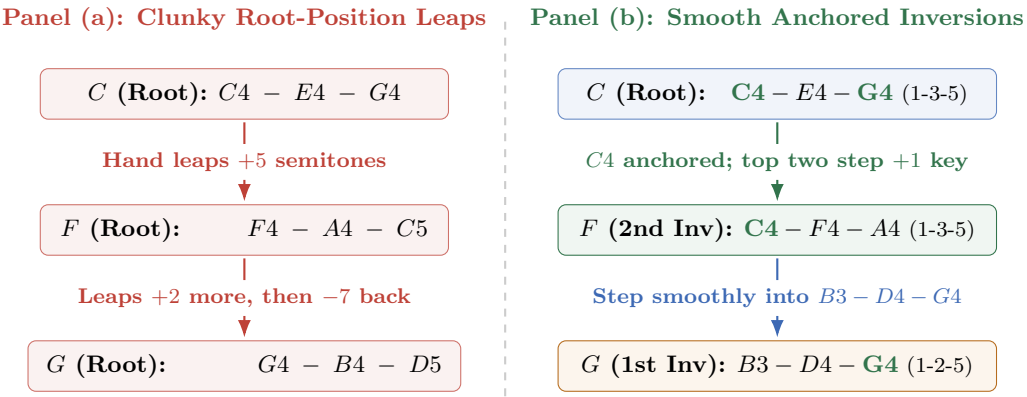


Figure 4.2: Left: Jumping between root-position triads forces your right hand to leap across a nine-key span ($C4$ to $D5$), breaking eye contact with the music. Right: Using anchored inversions keeps all three chords inside one calm six-key zone ($B3$ to $A4$) around Middle C.

4.3 Two-Hand Piano Accompaniment That Sounds Full Today

If your right hand is playing inverted chords such as $C4 - F4 - A4$ (where C, not F, sits at the bottom of your right hand), why does the listener still hear a clear F major chord rather than a blurred cluster? Because your **left hand** plays a low $F2$ or $F3$ in the bass, establishing the true root for the whole room.

Common trap

Beginners often try to play the exact same closed three-note triad in *both* hands at once — putting $C2 - E2 - G2$ down in the low bass alongside $C4 - E4 - G4$ above. Try pressing $C2 - E2 - G2$ low on a real piano and listen closely: it sounds thick and muddy. Thick low piano strings produce strong overtones that clash when thirds sit closer than an octave below $C3$ (pianists call this the **low-interval mud zone**). Keep closed three-note chords above $C3$, and give the low bass wide, open intervals: a single bass root, a fifth ($C2 - G2$), or an octave

$(C2 - C3)$.

Figure 4.3 shows the clean studio spacing used by ballad, pop, gospel, and classical pianists alike: left hand anchors wide roots or fifths down in the $C2$–$C3$ bass register, while right hand voices smooth three-note triads right around Middle $C4$.

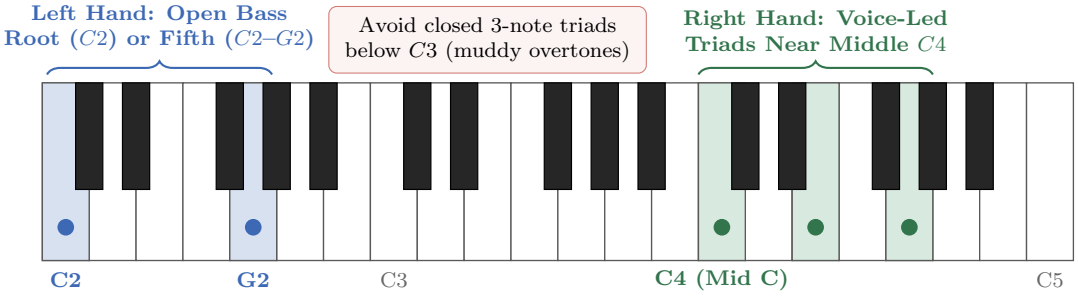


Figure 4.3: Balanced two-hand spacing across three octaves ($C2$ to $C5$). Left hand plays clean single roots, fifths, or octaves in the bass while right hand keeps three-note inversions inside a warm zone near Middle $C4$.

Once your hands sit in those two zones, three rhythmic accompaniment patterns bring any chord sheet to life immediately:

1. **The Steady Pulse (Singer-Songwriter Quarter Groove):** Hold a single bass note in your left hand for four steady counts while your right hand presses its three-note chord gently on every beat $(1, 2, 3, 4)$.
2. **The Rolling Wave (Broken-Chord Ballad):** Hold the left-hand bass note on beat 1, while your right hand unpacks its three notes one finger at a time: Bottom $\rightarrow$ Middle $\rightarrow$ Top $\rightarrow$ Middle across the bar.
3. **The Pop Bounce (Together – Right – Left – Right):** Press left-hand pinky ($C2$) and right-hand chord *together* on beat 1, right-hand chord alone on beat 2, left-hand thumb ($G2$) on beat 3, and right-hand chord on beat 4.

4.4 Slash Chords Demystified: Walking Basslines Made Easy

Open almost any songbook or guitar-and-piano chord sheet and you will meet symbols with a forward slash, such as C/E, G/B, or $D/F\sharp$. Beginners sometimes mistake these for fractions or two chords mashed together. The rule could not be simpler:

$$\underbrace{\text{Chord Name}}_{\text{Right-hand triad}} \Big/ \underbrace{\text{Bass Note}}_{\text{Left-hand single note}} .$$

Thus G/B literally reads "*G major chord over a B bass note*": your right hand plays any smooth G major chord around Middle C, while your left hand presses the single note B down in the bass.

Why do songwriters write G/B instead of plain G? Look at the bass notes across the classic opening line $C \rightarrow G/B \rightarrow Am \rightarrow Am/G \rightarrow F$:

$$\text{Left-hand bass:} \qquad C3 \rightarrow B2 \rightarrow A2 \rightarrow G2 \rightarrow F2.$$

Instead of hopping wildly from C down to G and up to A, the left hand walks smoothly step by step down the white keys like a cello player, turning five ordinary chords into an unforgettable melodic bassline.

Work it yourself

Eyes-closed progression drill: Place your left hand over $C3, B2, A2, G2, F2$ (fingers $1-2-3-4-5$ starting with thumb on $C3$) and your right hand on $C4-E4-G4$. Play four slow counts per chord across:

$$C \;\to\; G/B \;\to\; Am \;\to\; F \;\to\; C,$$

using smooth right-hand inversions ($C4$–$E4$–$G4$ $\to$ $B3$–$D4$–$G4$ $\to$ $C4$–$E4$–$A4$ $\to$ $C4$–$F4$–$A4$ $\to$ $C4$–$E4$–$G4$). After two slow rounds looking at your fingers, close your eyes and feel every shared anchor note guide your hand without looking.

Exercises

Exercise 4.1. Spell out all three positions (root position, first inversion, and second inversion) of (a) F major (F, A, C) and (b) G major (G, B, D). List the recommended right-hand fingering ($1-3-5$ or $1-2-5$) for each shape.

Exercise 4.2. Use the Gap Detector (find the wider 5-semitone fourth gap; the note immediately above it is the root) to identify the chord name and inversion of: (a) $F\sharp 4-A4-D5$, (b) $E\flat 4-A\flat 4-C5$, and (c) $B\flat 3-E\flat 4-G4$.

Exercise 4.3. Starting from root-position G major ($G4-B4-D5$), apply the Anchor-Note Law to find the closest smooth right-hand inversions for the four-chord progression $G \to D \to Em \to C$ ($I-V-vi-IV$ in G major). Write down the exact three notes and shared anchor note at each step.

Exercise 4.4. What should your left hand and right hand each play when a song sheet shows (a) $D/F\sharp$ between G and Em, and (b) C/G right before $G7 \to C$? Trace the three left-hand bass notes of $G \to D/F\sharp \to Em$ to see why songwriters love $D/F\sharp$.

Part III

Time, Rhythm, and Reading Sheet Music Effortlessly

Chapter 5

Beats, Timing, and Reading Sheet Music Without Counting Letters

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Play all the right notes of *Happy Birthday* with random five-second pauses between them and nobody in the room will recognize the tune. Now tap the rhythm of *Happy Birthday* on a wooden table with a single pencil — zero pitches at all — and everyone recognizes it within two seconds. Pitch gives music its colors, while **time** gives music its skeleton and forward breath.

In this final chapter we connect steady timing, two-hand coordination, and effortless sheet-music reading into one calm set of habits. Just as we replaced chord memorisation with visual hand shapes, we will replace slow letter-by-letter staff counting (*Every Good Boy...*) with five symmetric landmarks and instant interval shapes.

5.1 Pulse, Bars, and Why $\frac{3}{4}$ Feels Different From $\frac{6}{8}$

Two words get mixed up constantly in everyday speech, yet separating them clears up half of rhythm right away:

- **The Pulse (Beat):** The steady, unchanging heartbeat underneath a song — where your foot taps naturally on the floor, measured in **beats per minute (BPM)**. At 60 BPM, one beat clicks once per second like a wall clock; at 96 BPM, it walks at an unhurried stroll.
- **The Rhythm:** The actual pattern of long and short notes played *on top of* that steady pulse. Notes can stretch across four beats or dart four times inside a single beat, while the underlying floor tap stays rock steady.

Western music slices note lengths by clean powers of two. Inside standard four-beat time:

- A **whole note** (an open hollow oval) rings for 4 **beats**.
- A **half note** (hollow oval with a vertical stem) rings for 2 **beats**.
- A **quarter note** (solid filled oval with a stem) rings for 1 **beat**.
- An **eighth note** (filled oval with a stem and 1 flag or beam) rings for $\frac{1}{2}$ **beat** (2 per beat).

- A **sixteenth note** (filled oval with 2 flags or beams) rings for $\frac{1}{4}$ **beat** (4 per beat).

Vertical **bar lines** group beats into equal boxes called **bars** (or **measures**). Two stacked numbers at the start of the line — the **time signature** — tell you how each bar is counted:

- **4/4 (Common time):** Four quarter-note beats per bar (**1** − 2 − 3 − 4). This is the home groove of pop, rock, ballads, soul, and classical marches.
- **3/4 (Simple triple — Waltz):** Three quarter-note beats per bar (**1** − 2 − 3). Split into eighth notes, its six halves form **three pairs** (**1**-& **2**-& **3**-&), like a graceful three-step waltz.
- **6/8 (Compound duple — Sway or Jig):** In arithmetic class, $\frac{3}{4} = \frac{6}{8}$; in music, they feel completely different. A bar of $\frac{6}{8}$ groups six eighth notes into **two groups of three** (**1**-2-3 **4**-5-6). Your foot taps only *twice* per bar (on **1** and **4**), rocking gently like a boat or an Irish jig.

Formal

Inside one bar of $\frac{4}{4}$ time, the **beat subdivision grid** aligns every rhythmic duration onto an eight-slot eighth-note ruler (**1** + **2** + **3** + **4** +, spoken "one and two and three and four and") or a sixteen-slot sixteenth-note ruler (**1 e** + **a** ...):

Note Value	Beats in $\frac{4}{4}$	Eighth Slots	Sixteenth Slots	Spoken Count Example
Whole note	4	8	16	Hold across **1** + 2 + 3 + 4+
Dotted half note	3	6	12	Hold across **1** + 2 + 3+
Half note	2	4	8	Hold across **1** + 2+
Dotted quarter note	1.5	3	6	Hold **1** + 2, next note on +
Quarter note	1	2	4	Hold **1**+
Eighth note	0.5	1	2	Strike on **1** and on +
Sixteenth note	0.25	0.5	1	**1** *e* + *a*

5.2 The Subdivision Cure for Dotted Notes, Ties, and Two Hands

Why do almost all beginners rush forward on long half notes, slow down on sixteenth notes, and guess wildly on dotted notes? Because they count only the big beat numbers $(1, 2, 3, 4)$, leaving wide, unmeasured silence between beats.

The cure used by studio drummers and concert pianists is **subdividing out loud**. Speak the smallest active grid unit steadily under your breath ("1 and 2 and 3 and 4 and"). Figure 5.1 shows how spoken subdivision clears up three classic rhythm hurdles:

1. **Dotted notes (×1.5 rule):** A small dot right after any notehead adds **half of that note's own length** $(1 + \frac{1}{2} = \frac{3}{2})$. A **dotted quarter note** lasts $1 + 0.5 = 1.5$ beats, or **three eighth-note slots**. Never try to guess "one and a half seconds" in mid-air. Say **1** − and − **2** while holding the dotted quarter down. Then strike the following eighth note crisply on **"and"** right after 2, landing squarely on beat 3.
2. **Ties and syncopation:** A curved line joining two notes of the *exact same pitch* is a **tie**. Hold the key down across both durations without striking it a second time. Suppose you strike a chord on an upbeat (such as the "and" of beat 2) and tie it across beat 3. The accent shifts off

the main beat, creating **syncopation** — the forward bounce in ragtime, jazz, soul, and pop piano.

3. **Two-hand coordination as one drum pattern:** Your brain does not run two separate stopwatches for your left and right hands. Instead, it merges both hands into **one sequence of four simple actions**: **T** (*Together*), **R** (*Right* alone), **L** (*Left* alone), or hold. Whenever a two-hand bar feels tangled, pencil `T`, `R`, and `L` above the "1 + 2 + 3 + 4+" slots. Tap that combined pattern slowly on your knees before touching the keys.

Pause

Pause and test a tie versus a slur: A curved line between two *identical* pitches is a **tie** (hold, do not replay). A curved line arching over *different* pitches is a **slur**, which asks you to connect those notes smoothly (*legato*) without choppy gaps between fingers.

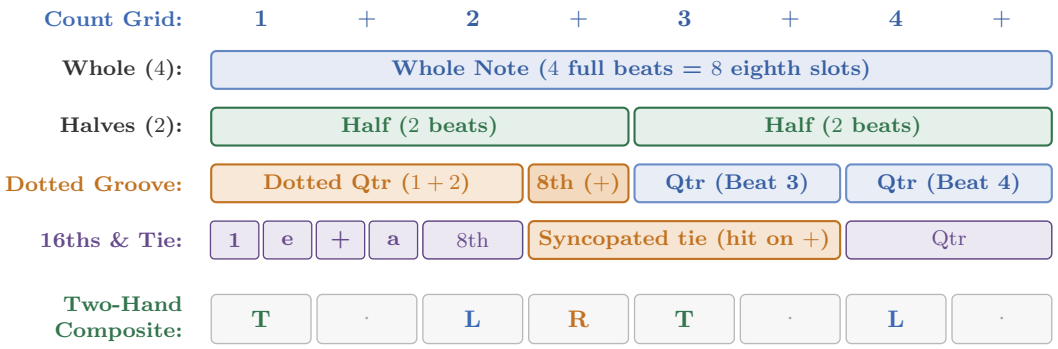


Figure 5.1: Proportional $\frac{4}{4}$ beat subdivision ruler (**1** + **2** + **3** + **4**+). A dotted quarter note fills three eighth-note slots (1 + 2), placing the next eighth note cleanly on the + of beat 2. Bottom row: Right Hand (dotted quarter + eighth + half) plus Left Hand (steady quarters) merges into one knee-tap pattern: `T - hold - L - R - T - hold - L - hold`.

5.3 The Grand Staff Is One Eleven-Line Mirror Centered on Middle *C*

Open a page of piano music and you see two five-line ladders braced together on the left: the upper **Treble Staff** (usually played by the right hand) and the lower **Bass Staff** (usually played by the left hand). Together they form the **Grand Staff**.

Traditional lessons teach beginners four separate mnemonic rhymes for the two staves (*Every Good Boy...*, *FACE*, *Good Boys Do Fine Always*, *All Cows Eat Grass*). That makes Treble and Bass look like two unrelated foreign alphabets, forcing you to pause and recite words in your head on every single note.

Look at Figure 5.2 to see what the Grand Staff actually is. It forms **one continuous eleven-line staircase of white piano keys**, stepping up one white key from each line to its touching space. Eleven solid lines in a row would blur before your eyes. Music printers solved that by **erasing the middle (sixth) line** — **Middle** *C* (*C*4) — drawing a tiny short dash (a **ledger line**) only when Middle *C* appears. Step one space below Middle *C* (*B*3) and you touch the top Bass line (*A*3). Step

one space above Middle C ($D4$) and you touch the bottom Treble line ($E4$).

Better still, the two clef signs act as crosshairs pointing to symmetric mirror landmarks around Middle C:

1. **Center Axis — Middle C ($C4$):** Sits on its single short ledger line directly between Bass and Treble.
2. **Up a Fifth (+4 staff steps) — Treble $G4$:** The inner spiral of the **Treble (G) Clef** wraps around the second line from the bottom, marking $G4$.
3. **Down a Fifth (−4 staff steps) — Bass $F3$:** The two dots of the **Bass (F) Clef** straddle the second line from the top, marking $F3$.
4. **Symmetric C Spaces ($C5$ and $C3$):** Space 3 from the bottom of Treble is $C5$, while space 3 from the top of Bass is $C3$. Two ledger lines outside either staff sit **High** $C6$ (top) and **Low** $C2$ (bottom).

Intuition

Pause and appreciate that mirror symmetry around Middle $C4$:

$$C2 \longleftrightarrow C3 \longleftrightarrow F3\ (F\text{-clef}) \longleftrightarrow \mathbf{C4} \longleftrightarrow G4\ (G\text{-clef}) \longleftrightarrow C5 \longleftrightarrow C6.$$

Every single key on the central four octaves of the piano sits at most **two steps away** from one of those landmarks. You never have to count up from the bottom line again.

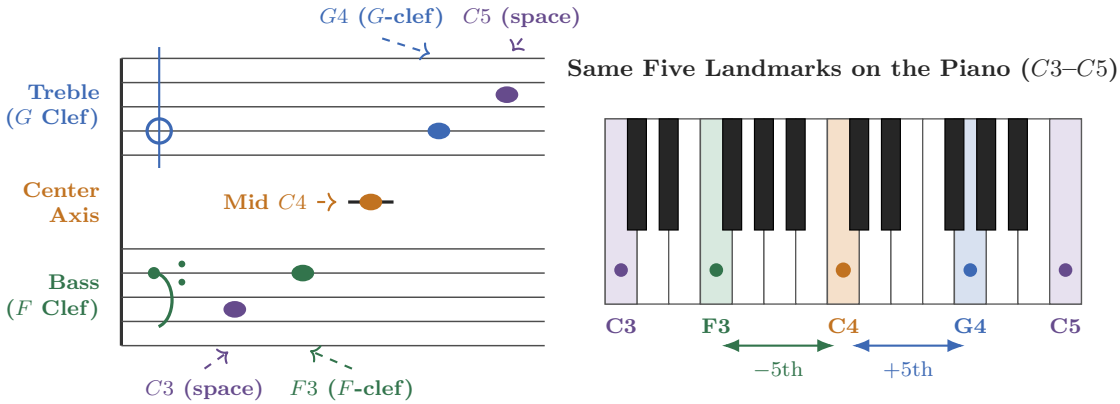


Figure 5.2: The Grand Staff revealed as an eleven-line mirror centered on Middle C ($C4$). Stepping up a fifth (+4 staff steps) lands on $G4$ (wrapped by the Treble G-clef spiral); stepping down a fifth (−4 staff steps) lands on $F3$ (framed between the two Bass F-clef dots); symmetric inner spaces hold $C3$ and $C5$.

5.4 Sight-Reading by Intervals and Chord Shapes

Fluent sight-readers almost never recite letter names in their heads after finding the first note of a phrase. Once your finger rests on that starting key, read **distances (intervals)** horizontally and **shapes** vertically, as in Figure 5.3:

- **2nd (Step — next scale key, next finger):** Moves from a *Line to its touching Space*, or *Space to touching Line*.
- **3rd (Skip — skip 1 key and 1 finger, such as $1 \to 3$):** Moves from a *Line to the next Line*, or *Space to the next Space*.
- **4th vs. 5th (Line/Space Parity):** Odd intervals (3rd, 5th, 7th) stay **same-to-same**: a **5th** jumps two lines away, matching $1 \to 5$. Even intervals (2nd, 4th, 6th, octave) always flip **Line-to-Space** or **Space-to-Line**.

Best of all, the three chord inversions from Chapter 4 appear as three clear silhouettes on the staff:

1. **Root-position triad ($1-3-5$):** Three noteheads stacked like a neat three-ball **snowman** (Line–Line–Line or Space–Space–Space) with zero gaps.
2. **First inversion ($1-2-5$):** Two noteheads snug at the bottom plus an open **gap on top** (top note is the root).
3. **Second inversion ($1-3-5$):** An open **gap at the bottom** plus two noteheads snug on top (middle note is the root).

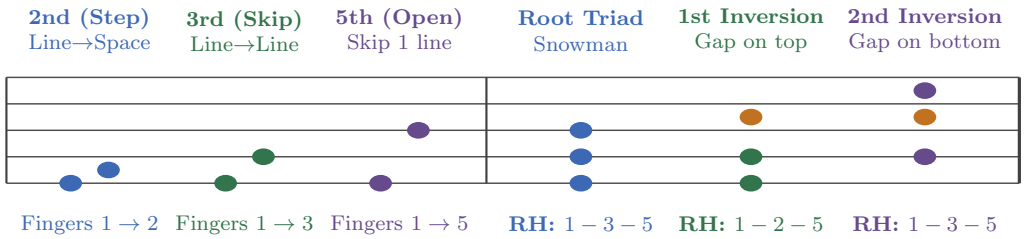


Figure 5.3: Sight-reading shapes on a five-line staff without naming every letter. Left: Steps (2nds) toggle line-to-space; skips (3rds) and 5ths stay line-to-line or space-to-space. Right: Root-position triads stack into an even snowman ($1-3-5$), first inversion opens a gap at the top ($1-2-5$, orange root on top), and second inversion opens a gap at the bottom ($1-3-5$, orange root in the middle).

5.5 Five Everyday Habits for Calm, Musical Sight-Reading

Bring all five chapters together whenever you open a new piece of music at the piano:

1. **Take a ten-second pre-flight scan before beat 1:** Check the **key signature** first so you know which black keys stay active. Next check the **time signature** ($\frac{4}{4}$, $\frac{3}{4}$, or $\frac{6}{8}$), and rest both hands over their starting landmarks before playing.
2. **Keep your eyes on the page, half a beat ahead of your hands:** Every time you glance down at your fingers, you lose your place on the staff. Trust the raised black-key groups and smooth anchor-note inversions under your fingertips so your eyes glide steadily ahead.
3. **Protect the pulse above individual pitches:** When you clip a wrong note, *never stop, back up, and replay the beat.* Listeners forget a slipped pitch within half a second, whereas stopping breaks the music. Let the missed note go and land calmly on the next downbeat.

4. **Read vertical stacks from the bottom up:** Just as a building stands on its foundation, spot the lowest left-hand bass note first. Then grab the right-hand chord silhouette (snowman, top gap, or bottom gap) above it.
5. **Practice four-bar phrases hands-together at half speed:** Beginners often drill each hand fast ten times alone, then stall when combining them. Playing four bars *hands-together* at 50% speed wires the unified `T / R / L` rhythm right from the start.

Common trap

Avoid starting a new piece at full concert speed on bar 1 and crashing into a wall on bar 3 where sixteen notes appear. Always choose your practice speed so that the *busiest* bar in the line feels calm and unhurried. Speed arrives naturally once your fingers trust the path.

Focus

Look back at how far we have come from the intimidating posters in the Preface. Twelve repeating keys ($\mathbb{Z}_{12}$) gave us C and F by touch, and one symmetric $W - W - H$ half-scale unlocked every key. One $4 + 3$ / $3 + 4$ frame and three visual profiles built every major and minor chord, while inversions kept chords inside a six-key zone. Finally, spoken subdivisions and five mirror landmarks turned sheet music into a direct map of your hands. Play slowly, listen closely, and enjoy the quiet craft of the piano.

Exercises

Exercise 5.1. Write out the eight eighth-note slots `1 + 2 + 3 + 4 +` across one bar of $\frac{4}{4}$ time. Mark `T` (Together), `R` (Right), `L` (Left), or `-` (hold) on each slot when the Left Hand plays half notes on beats 1 and 3. Meanwhile, the Right Hand plays a quarter note (beat 1), two eighth notes (beat 2), a dotted quarter (beat 3), and an eighth note (the `+` of beat 4).

Exercise 5.2. Explain in one sentence why six eighth notes in $\frac{3}{4}$ time and six eighth notes in $\frac{6}{8}$ time make your foot tap differently, even though $\frac{3}{4} = \frac{6}{8}$.

Exercise 5.3. Use the five Grand Staff mirror landmarks ($C3$, $F3$, Middle $C4$, $G4$, and $C5$) without counting from the bottom line. Name the note sitting (a) one line *above* $G4$, (b) one space *below* Middle $C4$, and (c) one line *below* $F3$.

Exercise 5.4. On the Treble Staff, three stacked noteheads sit on line 1 ($E4$), space 2 ($A4$), and space 3 ($C5$). Which silhouette (snowman, gap on top, or gap on bottom) does this chord form? Give its best right-hand fingering and its root name.